



Grade 11 Geography

5. Impacts of Development on the Environment

Learning Guide

1. Overview

Development can improve people's lives and grow economies, but it can also place pressure on the **natural environment**.

The following are some major environmental impacts of development:

- **Deforestation**
- **Pollution**
- **Climate change**
- **Resource depletion**
- **Loss of biodiversity**
- **Urbanisation**
- **Waste generation**

★ **Key idea:** Development can create economic benefits, but if it is poorly managed it may damage ecosystems, reduce natural resources, and increase pollution.

2. Deforestation

- **Deforestation** is the loss or clearing of forests.
- Forests may be removed for:
 - Agriculture
 - Urban development

Effects of Deforestation

- Loss of animal and plant habitats.
- Reduced biodiversity.
- Increased carbon emissions.
- Soil erosion.
- Less oxygen (O_2).
- More carbon dioxide (CO_2).
- Loss of natural windbreaks.

Sustainable Solution

- **Trees must be replanted.**

★ **Key idea:** Cutting forests removes vegetation that protects soil, supports habitats, and helps regulate gases in the atmosphere.

3. Pollution

Development can cause pollution of:

- **Air**
- **Water**
- **Soil**

Sources

- Industrial activities.
- Mining.
- Waste disposal.
- Chemicals.
- Sewage.
- Household products.

Possible Effects

- Contaminated rivers.
 - Contaminated soil.
 - Air pollution.
 - Damage to ecosystems.
 - Harm to human and animal health.
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4. Climate Change

- Development can increase the release of **greenhouse gases**.
- Greenhouse gas emissions increase **global temperatures**.

Possible Effects

- Melting ice caps.
- Rising sea levels.
- Heatwaves.
- Warming oceans.
- Extreme weather events.
- Species extinction.

 **Key idea:** Increased greenhouse gases contribute to global warming and wider environmental changes.

5. Resource Depletion

- **Resource depletion** occurs when natural resources are used faster than they can be replaced.
- Resources affected include:
 - Water
 - Minerals
 - Fossil fuels

Result

- Essential resources become **scarce**.

 **Key idea:** Overuse today can reduce the availability of resources in the future.

6. Loss of Biodiversity

- **Biodiversity** refers to the variety of living organisms.
- Development can destroy habitats and lead to **species extinction**.

Causes include:-

- Deforestation.
- Mining.
- Urbanisation.
- Overfishing.
- Pollution.
- Destruction of sensitive areas.

 **Key idea:** When habitats are destroyed, species may lose food, shelter, breeding areas, and eventually disappear.

7. Urbanisation

- **Urbanisation** causes cities and settlements to expand.
- This changes **land use**.
- Urban development can place greater pressure on:
 - Water
 - Land
 - Waste systems
 - Natural habitats

Possible Environmental Effects

- Wetlands may be drained.
- More landfills are needed.
- Informal settlements may expand.
- Sewage spills and litter may increase.

- Natural habitats may be destroyed.

 **Key idea:** Urban expansion can replace natural environments with buildings, roads, waste sites, and other infrastructure.

8. Waste Generation

- Development and increased consumption produce more:
 - Solid waste.
 - Plastics.
 - Household waste.
 - Industrial waste.

Effects

- Pollution of land.
- Pollution of oceans.
- Animals may eat or become tangled in plastic.
- Waste may contaminate soil and water.

 **Key idea:** Increased consumption usually leads to increased waste unless waste is properly managed.

9. Harvesting Resources: Agriculture

Poor farming methods may result in:

- **Overgrazing**
- Soil becoming **infertile**
- **Soil erosion**

Solutions for Sustainable Agriculture

- **Crop rotation**
- **Fertilisation**
- Education in **sustainable farming**

 **Key idea:** Farming must protect soil fertility while still producing food.

10. Mining and the Environment

Mining, especially **open-cast mining**, can cause major environmental damage.

Impacts

- Destruction of ecosystems.
- Formation of mine dumps.
- Chemicals from mining may drain into:
 - Underground water sources.
 - Rivers.
- Fine solid particles from open-cast mining may cause **atmospheric pollution**.

Solution

- Laws restricting and monitoring mining operations should be **strictly implemented**.

 **Key idea:** Mining may provide resources, but poor management can damage land, water, and air.

11. Overfishing

- **Overfishing** removes too many fish from the ocean.
- This disturbs:
 - The food chain.
 - The natural balance of ecosystems.

Effects

- Fewer fish in the ocean.
- Possible extinction of some species.

Solution

- Improve and strictly enforce:
 - Fishing laws.
 - Fishing quotas.

 **Key idea:** Fish populations need time to recover, so harvesting must be controlled.

12. Harvesting Energy

Coal

- Coal increases:
 - **CO₂**
 - Sulphur pollution.

Wind Turbines

- Wind turbines may harm:
 - Bats.
 - Birds.

Sustainable Energy Solutions

Use renewable energy sources such as:

- **Hydroelectric power**
- **Solar energy**

 **Key idea:** Energy production can have environmental costs, so cleaner energy sources should be promoted.



13. Manufacturing and Industry

Manufacturing can cause:

Air Pollution

- Combustion releases:
 - CO₂.
 - Other greenhouse gases.
- This contributes to **global warming**.

Water Pollution

- Industrial waste may reach rivers.

Solutions

- Fit **filters to chimney stacks** to:
 - Reduce harmful gases.
 - Prevent solid particles from entering the atmosphere.
- Introduce and enforce **strict regulations** controlling:
 - Factory waste.
 - Greenhouse gas emissions.

 **Key idea:** Industrial growth should be controlled so that production does not severely damage air and water quality.



14. Tourism

Tourism can also place pressure on the environment.

Impacts

- Vegetation may be cleared to build hotels.
- Traffic increases.
- Litter increases.
- Coral reefs may be damaged.

Solution

- **Limit access to sensitive areas.**

 **Key idea:** Tourism should be managed so that visitors do not destroy the environments they come to enjoy.



15. Construction of Infrastructure and Housing

Urban development and infrastructure can produce several environmental problems.

A. Dam Building

Impact

- Dams may change or destroy:
 - Aquatic life.
 - Fish migration in rivers.

Solution

- Conduct an **Environmental Impact Assessment (EIA)** before dams are built.

📌 **Key idea:** *The environmental consequences of major projects should be investigated before construction begins.*

B. Landfills and Nuclear Waste

Impact

- Many landfills are built around urban areas.
- Nuclear waste must also be buried.

Solution

- Introduce strict laws on:
 - Waste management.
 - Recycling.

💧 16. Eutrophication

Eutrophication may be caused by:

Sewage Leaks

- Old sewage treatment plants may not be properly maintained.
- Sewage may leak into water systems.

Fertilisers

- Commercial farmers use fertilisers.
- After rain, fertilisers may run into:
 - Rivers.
 - Dams.

Solution

- Improve infrastructure.
- Strengthen and enforce **by-laws on dumping**.

📌 **Key idea:**

Poor waste and water management can contaminate rivers and dams.

🌿 17. Draining of Wetlands

- Wetlands may be drained for **urban development**.
- This has resulted in **increased flooding**.

Solution

- Wetlands must be **preserved in town planning**.

📌 **Key idea:** *Wetlands perform important natural functions and should not simply be removed for development.*

🏠 18. Informal Settlements

- The growth of informal settlements may be accompanied by:
 - Litter.
 - Sewage spills.

Solution

- Municipalities must control **sprawling development**.

🦌 19. Habitat Destruction

- Development may destroy natural habitats.
- **Knysna seashore** as an example.

Solutions

- Open more **conservation areas**.
- Encourage **wildlife corridors**.

📌 **Key idea:** *Protected spaces and wildlife corridors help maintain habitats as development expands.*

🛒 20. A Culture of Thoughtless Consumption

Development can also increase consumption.

Overconsumption

- Buying too many goods such as:
 - Food.
 - Gadgets.
 - Plastic toys.
- Produces large amounts of waste.

Solution

- **Awareness and education** are vital.
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21. Littering

- Non-biodegradable waste such as **plastic packets** damages the environment.
- Animals such as:
 - Seagulls.
 - Turtles.
 - Seals.
- May eat plastic or become tangled in it.

Solution

- **Do not litter.**
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22. Household Chemicals

- Soaps, detergents, and fabric softeners may:
 - Alter the **pH balance**.
 - Contaminate:
 - Soil.
 - Rivers.
 - Oceans.

Solution

- Use **biodegradable, environmentally friendly products**.
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23. Harmful Development vs Sustainable Solutions

Environmental Problem	Sustainable Solution
Poor farming	Crop rotation, fertilisation, sustainable farming education
Mining damage	Strict mining laws and monitoring
Overfishing	Fishing laws and quotas
Deforestation	Replant trees
Coal pollution	Renewable energy
Industrial air pollution	Filters on chimney stacks
Industrial waste	Strict factory regulations
Tourism pressure	Limit access to sensitive areas
Dam construction	Environmental Impact Assessments
Landfills	Waste management and recycling
Eutrophication	Improve infrastructure and dumping by-laws
Wetland destruction	Preserve wetlands
Urban sprawl	Municipal control
Habitat destruction	Conservation areas and wildlife corridors
Overconsumption	Awareness and education
Littering	Do not litter
Household chemical pollution	Use biodegradable products



24. Final Exam Summary

- Development can negatively affect the environment through:
 - Deforestation.
 - Pollution.
 - Climate change.
 - Resource depletion.
 - Loss of biodiversity.
 - Urbanisation.
 - Waste generation.
- Agriculture may cause:
 - Overgrazing.
 - Soil infertility.
 - Soil erosion.
- Mining may damage:
 - Ecosystems.
 - Water.
 - Air quality.
- Overfishing can disrupt food chains and reduce fish populations.
- Coal increases CO₂ and sulphur pollution.
- Industry may cause:
 - Air pollution.
 - Water pollution.
 - Greenhouse gas emissions.
- Tourism may damage vegetation and sensitive ecosystems.
- Urban development may affect:
 - Wetlands.
 - Rivers.
 - Habitats.
 - Waste systems.
- Sustainable development requires:
 - Better farming methods.
 - Strict environmental laws.
 - Renewable energy.
 - Pollution controls.
 - Recycling.
 - EIAs.
 - Conservation.
 - Education and awareness.

🔴 **Core idea:** Development should satisfy human needs while reducing damage to natural resources, ecosystems, and future generations.

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